

Quiz Master - V1: Project Report

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About Me

I am Aastha Kaushik, a diploma student, and I developed 'Quiz Master - V1.

Description

The "Quiz Master - V1" project is a web-based quiz management system. It allows users to attempt quizzes and track scores while administrators manage quiz content. The project requires user authentication, quiz attempt tracking, and a structured database design.

Technologies Used

- **Flask** (Web framework for handling routes and server-side logic)
- **Flask-SQLAlchemy** (ORM for database interaction)
- **SQLite** (Lightweight database for storing quiz data)
- **Jinja2** (Templating engine for dynamic HTML pages)
- **Flask-Login** (User authentication and session management)
- **Matplotlib** (Generating score visualization graphs)

DB Schema Design

The database consists of the following tables:

- **Users** (id, username, password, role)
- **Subjects** (id, subject_name)
- **Chapters** (id, chapter_name, subject_id)
- **Quizzes** (id, quiz_name, chapter_id, duration)
- **Questions** (id, question_text, quiz_id, option1, option2, option3, option4, correct_option)
- **Scores** (id, user_id, quiz_id, score, timestamp)

Design Rationale

- User authentication ensures **role-based access** (Admin/User).

- Relational structure allows **efficient quiz management**.
- The **scores table** tracks user performance over time.

API Design

- **User Authentication API:** Handles login and registration.
- **Quiz Retrieval API:** Fetches quizzes for users.
- **Score Submission API:** Stores user scores upon quiz completion.
- **Admin API:** Allows admin to create/manage quizzes.

A YAML file containing API definitions is submitted separately.

Architecture and Features

Project Organization

- **app.py** (Controllers / Views )
- **templates/** (Jinja2 Templates )
- **User Authentication** (admin_login.html, user_login.html, register.html )
- **Quiz Management** (manage_quizzes.html, create_quiz.html, edit_quiz.html )
- **Chapters & Subjects** (manage_chapters.html, add_subject.html, edit_chapter.html )
- **Admin & User Management** (admin_dashboard.html, manage_users.html )
- **Question Management** (manage_questions.html, add_question.html, edit_question.html )
- **Matplotlib**

Video Presentation

1. https://drive.google.com/file/d/1B6sjvooP5M_35qLIsM8_A8-Z1ZPCvcQv/view?usp=sharing

Conclusion

"Quiz Master - V1" is a **Flask-based quiz application** designed to help students prepare for exams through interactive quizzes and performance tracking. The project ensures a user-friendly experience with structured data management, authentication, and score visualization. Future improvements could include **leaderboards, time-based quizzes, and mobile optimization**.